Let $y = x - 2$. As $x \to \infty$, $y \to \infty$.

$$\lim_{x\to\infty}\left(\frac{x+2}{x-2}\right)^{x} = \lim_{y\to\infty}\left(1+\frac{4}{y}\right)^{y+2} = \lim_{y\to\infty}\left(1+\frac{4}{y}\right)^{y}\lim_{y\to\infty}\left(1+\frac{4}{y}\right)^{2}$$

$$= e^4(1)^2 = e^4 .$$

Example 9.34

Evaluate : $\lim\limits_{x\to\frac{\pi}{4}} \dfrac{4\sqrt{2}-(\cos x+\sin x)^5}{1-\sin 2x}$.

Solution

$$\frac{4\sqrt{2}-(\cos x+\sin x)^5}{1-\sin 2x} = \frac{2^{\frac{5}{2}}-\left[(\cos x+\sin x)^2\right]^{\frac{5}{2}}}{1-\sin 2x}$$

$$= \frac{2^{\frac{5}{2}}-(1+\sin 2x)^{\frac{5}{2}}}{2-(1+\sin 2x)}$$

$$= \frac{2^{\frac{5}{2}}-(1+\sin 2x)^{\frac{5}{2}}}{2-(1+\sin 2x)}$$

Therefore, $\lim\limits_{x\to\frac{\pi}{4}} \dfrac{2^{\frac{5}{2}}-[(\cos x+\sin x)^2]^{5/2}}{2-[1+\sin 2x]} = \lim\limits_{x\to\frac{\pi}{4}} \dfrac{2^{\frac{5}{2}}-\left[1+\sin 2x)\right]^{\frac{5}{2}}}{2-(1+\sin 2x)}$.

Take $y = 1 + \sin 2x$. As $x \to \dfrac{\pi}{4}$, $y \to 2$

$$= \lim_{y\to 2} \frac{2^{\frac{5}{2}}-y^{\frac{5}{2}}}{2-y}$$

$$= \frac{5}{2}.2^{\frac{5}{2}-1} = \frac{5}{2}\times 2^{\frac{3}{2}} = 5\sqrt{2} .$$

Example 9.35

Do the limits of following functions exist as $x \to 0$? State reasons for your answer.

(i) $\dfrac{\sin|x|}{x}$ (ii) $\dfrac{\sin x}{|x|}$ (iii) $\dfrac{x\lfloor x\rfloor}{\sin|x|}$ (iv) $\dfrac{\sin\left(x-\lfloor x\rfloor\right)}{x-\lfloor x\rfloor}$.

Solution

(i) $$f(x) = \begin{cases} \dfrac{\sin(-x)}{x} & \text{if } -1 < x < 0 \\ \dfrac{\sin x}{x} & \text{if } 0 < x < 1 \end{cases}$$

Therefore, $\lim\limits_{x\to 0^-} f(x) = -1$ and

$\lim\limits_{x\to 0^+} f(x) = +1$.

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